Section 1

AC-CIRCUIT + MACHINERY PRINCIPLE

1. The magnetic field

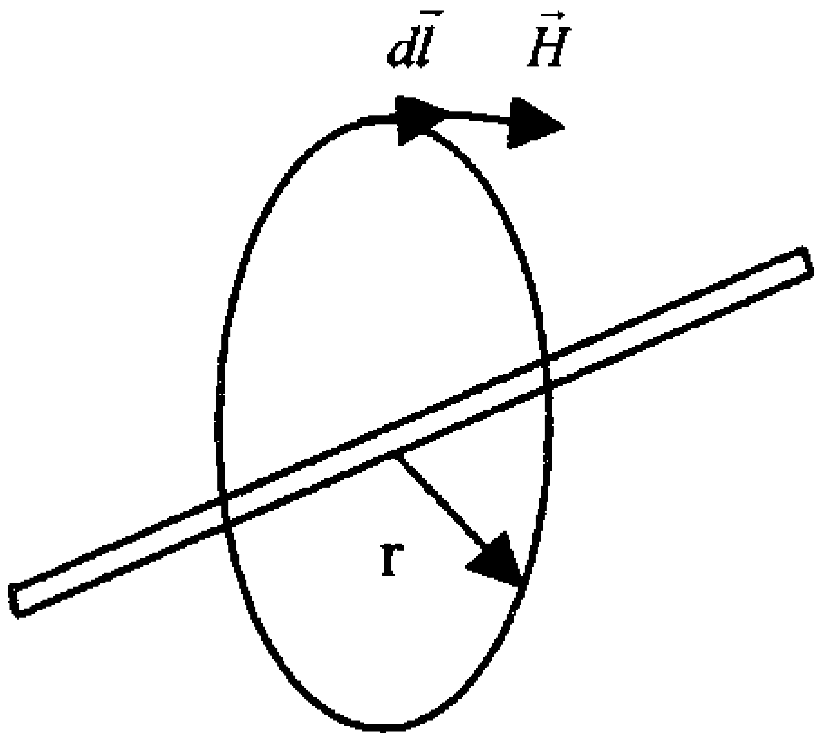
Magnetic fields are the fundamental mechanism by which energy is converted from one form to another in motors, generators, and transformers. Four basic principles describe how magnetic fields are used in these devices:

1. A current-carrying wire produces a magnetic field in the area around it.
2. A time changing magnetic field induces a voltage in a coil of wire if it passes through that coil. (This is the basis of transformer action).
3. A current-carrying wire in the presence of a magnetic field has a force induced on it. (This is the basis of motor action).
4. A moving wire in the presence of a magnetic field has a voltage induced in it. (This is the basis of generator action.)

a. Production of a magnetic field

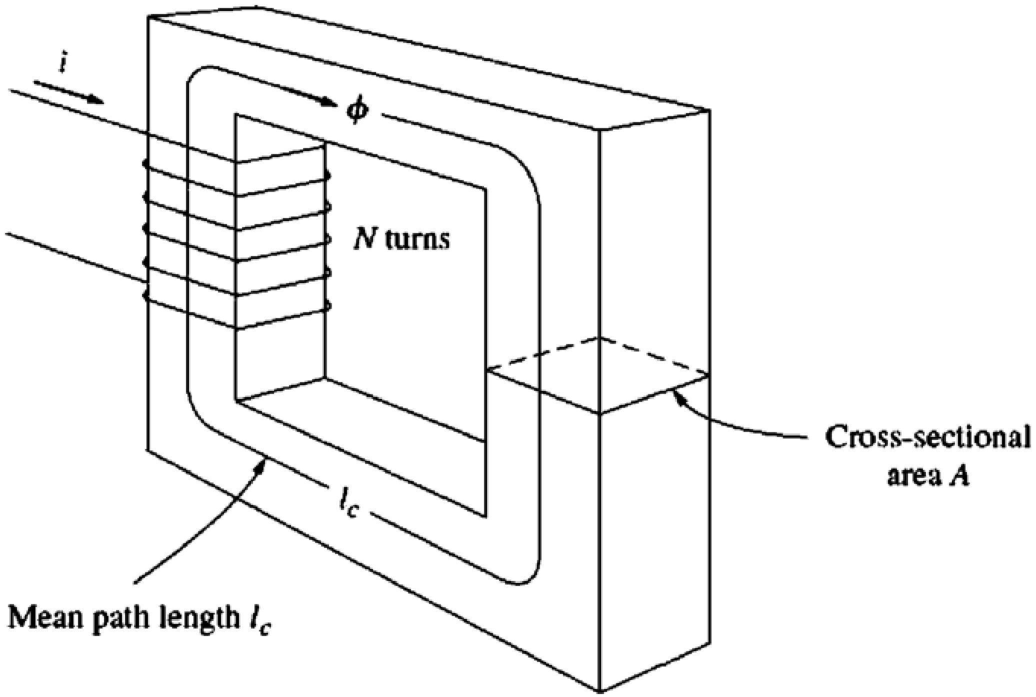
The basic law production of a magnetic field by a current is Ampere's law:

$$\oint \mathbf{H} \cdot d\mathbf{l} = I_{\text{net}}$$



Where *H* is the magnetic field intensity produced by the current I_{net} and *d*l is a differential element of length along the path of integration.

To better understand the meaning of this equation, it is helpful to apply it to the simple example:



- A rectangular core with a winding of *N* turns of wire wrapped about one leg of the core.
- If the core is composed of iron or certain other similar metals, essentially all the magnetic field produced by the current will remain in side the core, so the path of integration in Ampere's law is the mean path length of the core *lc*.

Ampere's law thus becomes:

$$Hl_c = Ni$$

$$H = \frac{Ni}{l_c}$$

The strength of the magnetic field flux produced in the core also depends on the material of the core. The relationship between the magnetic field intensity *H* and the resulting magnetic flux density *B* produced within a material is given by:

$$\mathbf{B} = \mu \mathbf{H}$$

$$B = \mu H = \frac{\mu N i}{l_c} \quad ; \quad \mu = \mu_r \mu_0 \quad ; \quad \mu_0 = 4\pi \times 10^{-7}\, H/m$$

where

$\mathbf{H}$ = magnetic field intensity

μ = magnetic *permeability* of material

$\mathbf{B}$ = resulting magnetic flux density produced

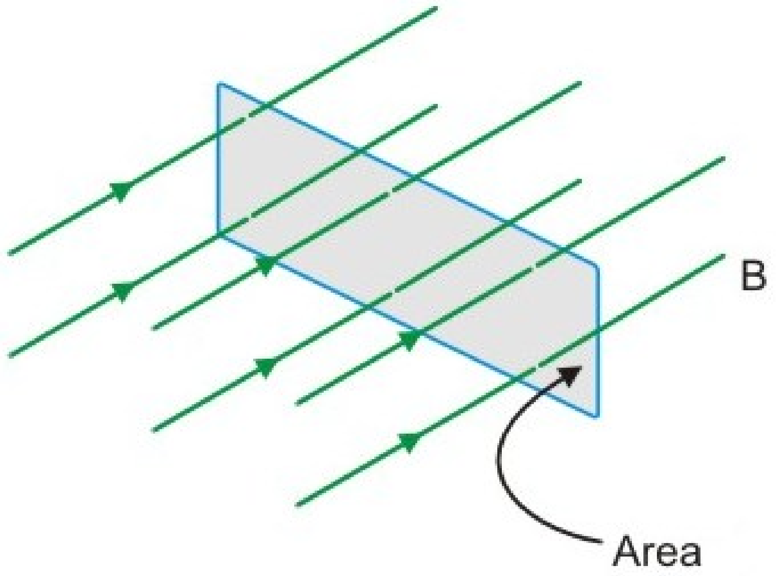
μ_r = Relative permeability

Now the total flux in a given area is given by:

$$\phi = \int_A \mathbf{B} \cdot d\mathbf{A}$$

$$\phi = BA$$

$$\phi = BA = \frac{\mu N i A}{l_c}$$



b. Magnetic Circuit

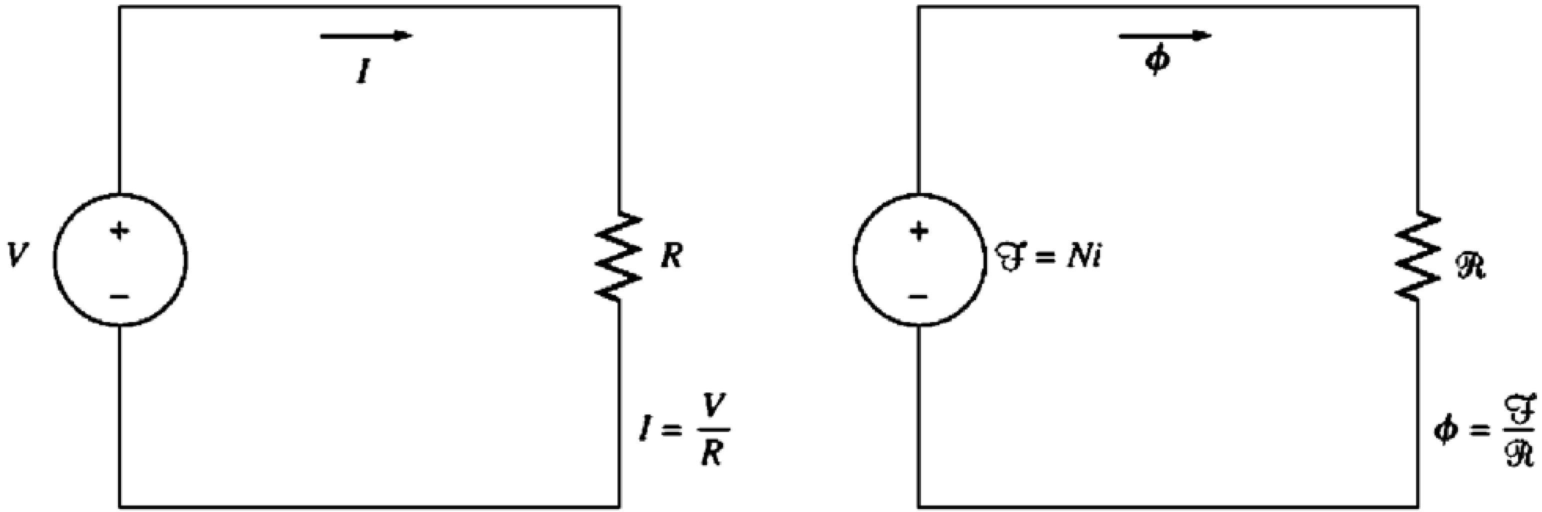
The voltage source *V* drives a current *I* around the circuit through a resistance *R* by Ohm's law.

$$V = IR$$

The magnetomotive force of the magnetic circuit is equal to the effective current now applied to the core, or:

$$\mathcal{F} = Ni$$

where $\mathcal{F}$ is the symbol for magnetomotive force, measured in ampere-turns.



In an electric circuit, the applied voltage causes a current to flow. Similarly, in a magnetic circuit, the applied magnetomotive force causes the flux to be produced.

$$\boxed{\mathcal{F} = \phi \mathcal{R}}$$

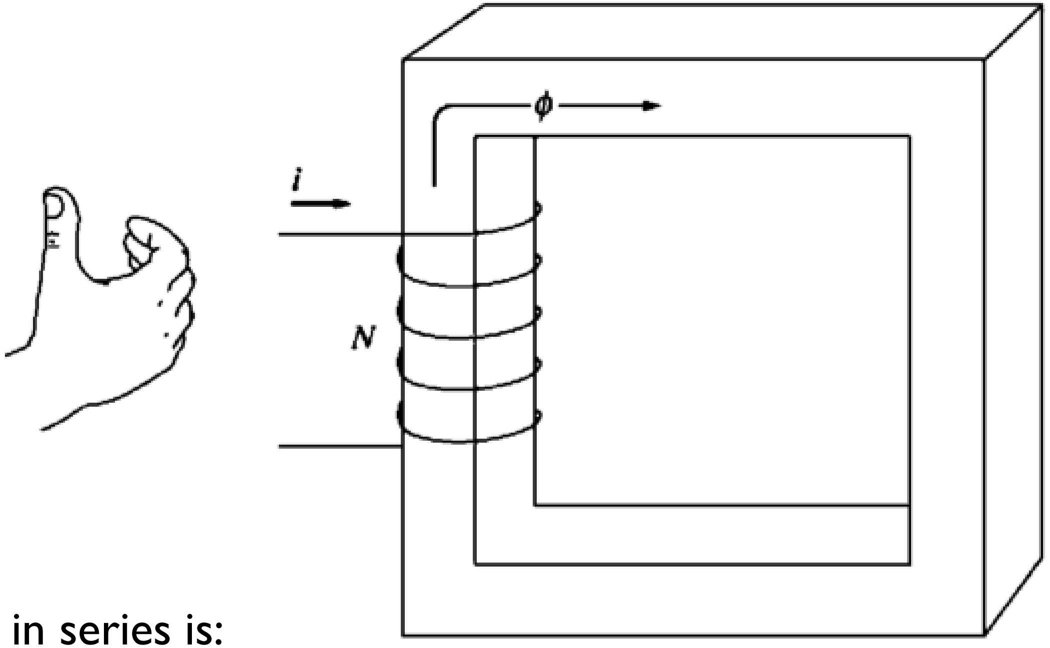
where

$\mathcal{F}$ = magnetomotive force of circuit

ϕ = flux of circuit

$\mathcal{R}$ = *reluctance* of circuit

$$\mathcal{R} = \frac{l_c}{\mu A}$$



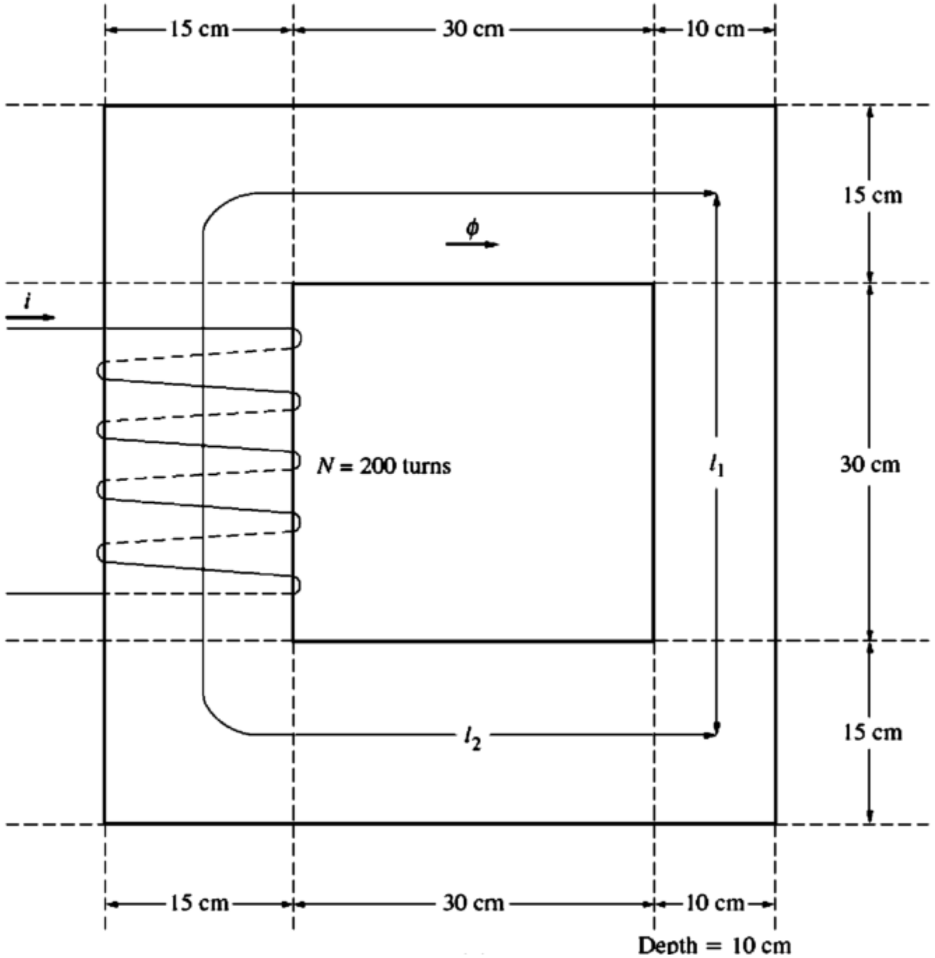
- The equivalent reluctance of a number of reluctances in series is:

$$\mathcal{R}_{eq} = \mathcal{R}_1 + \mathcal{R}_2 + \mathcal{R}_3 + \cdots$$

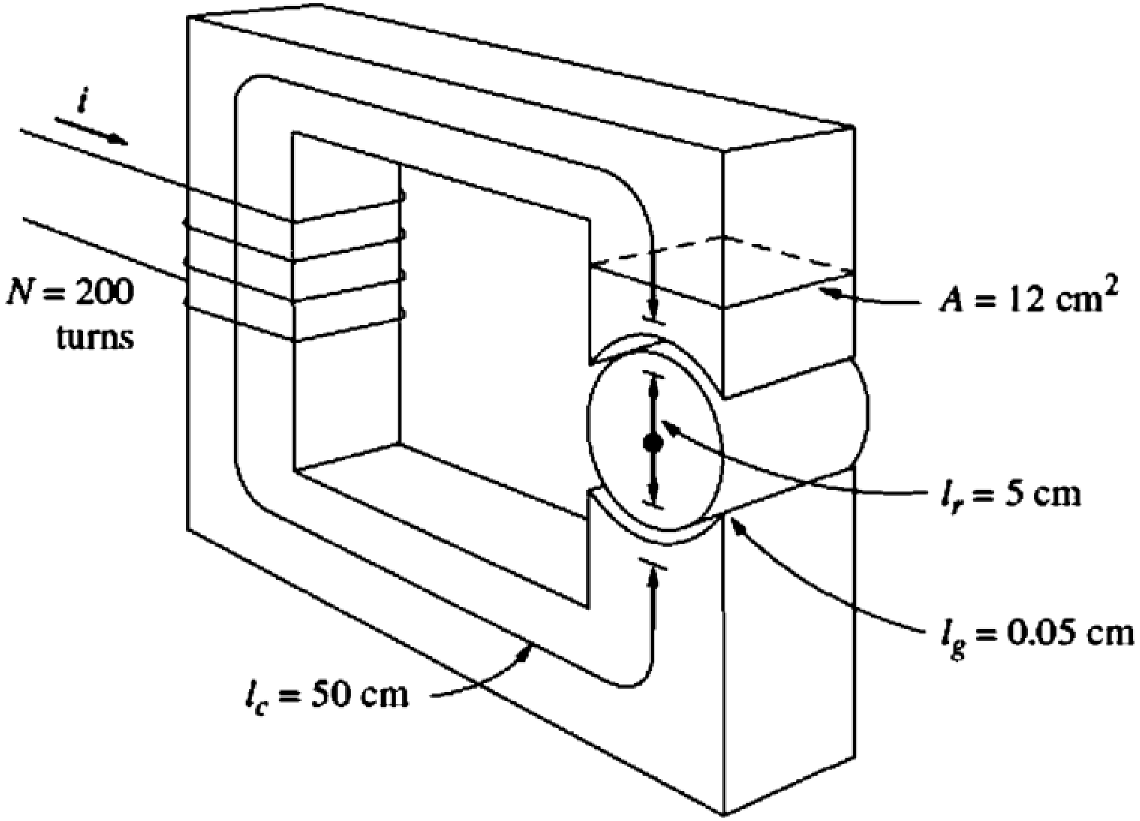
- Reluctances in parallel combine according to the equation:

$$\frac{1}{\mathcal{R}_{eq}} = \frac{1}{\mathcal{R}_1} + \frac{1}{\mathcal{R}_2} + \frac{1}{\mathcal{R}_3} + \cdots$$

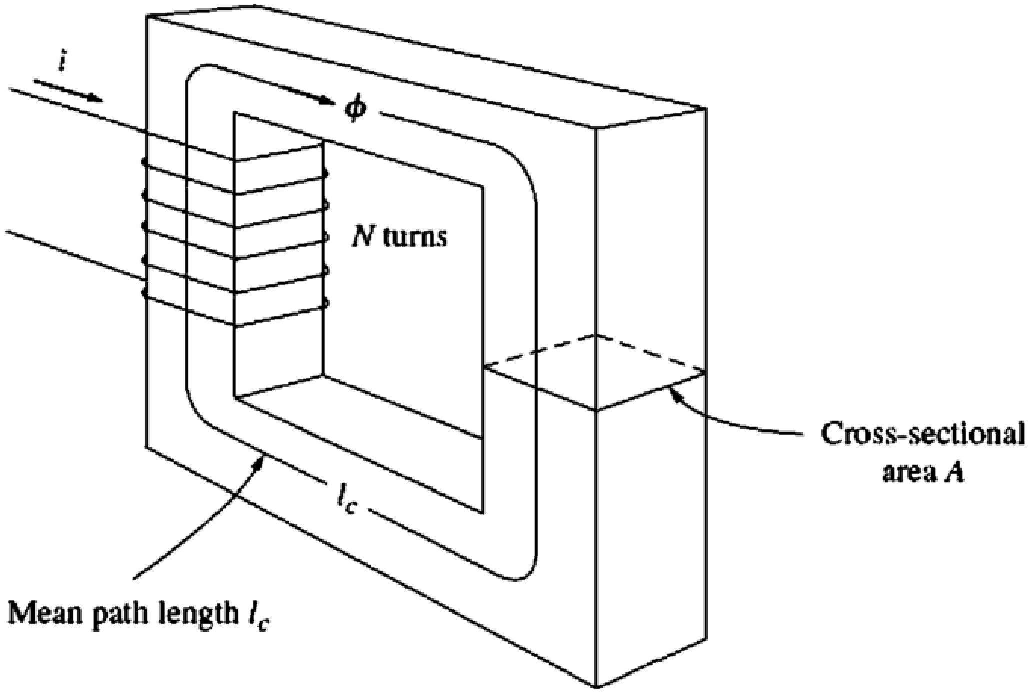
Example 1: A ferromagnetic core is shown in Figure below. Three sides of this core are of uniform width, while the fourth side is somewhat thinner. The depth of the core (into the page) is 10 cm, and the other dimensions are shown in the figure. There is a 200 turn coil wrapped around the left side of the core. Assuming relative permeability of μ_r of 2500. How much flux will be produced by a 1 A input current?



Example 2: In the following figure shows a simplified rotor and stator for a DC motor. The mean path length of the stator is 50 cm, and its cross-sectional area is 12 cm2 . The mean path length of the rotor is 5 cm, and its cross-sectional area also may be assumed to be 12 cm2 . Each air gap between the rotor and the stator is 0.05 cm wide, and the cross-sectional area of each air gap is 14 cm2 . The iron of the core has a relative permeability of 2000 and 1 for air, and there are 200 turns of wire on the core. If the current in the wire is adjusted to be 1A, what will the total flux?



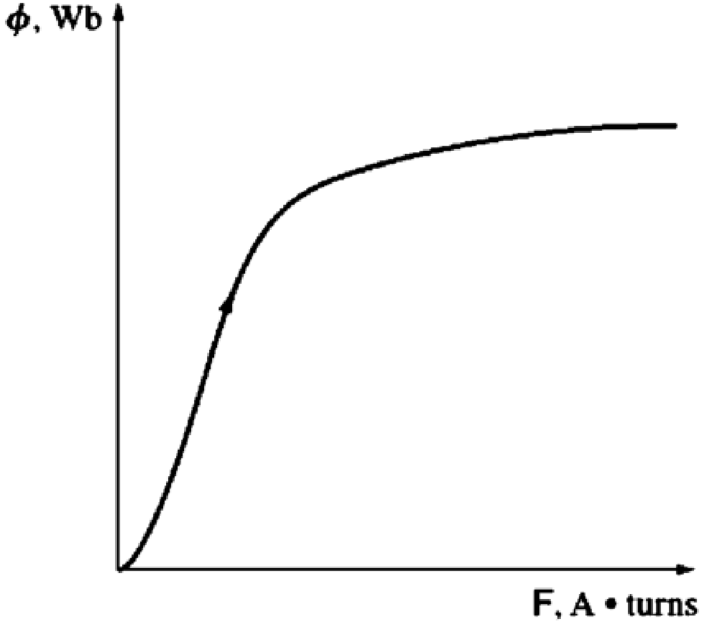
c. Magnetic behavior of ferromagnetic materials

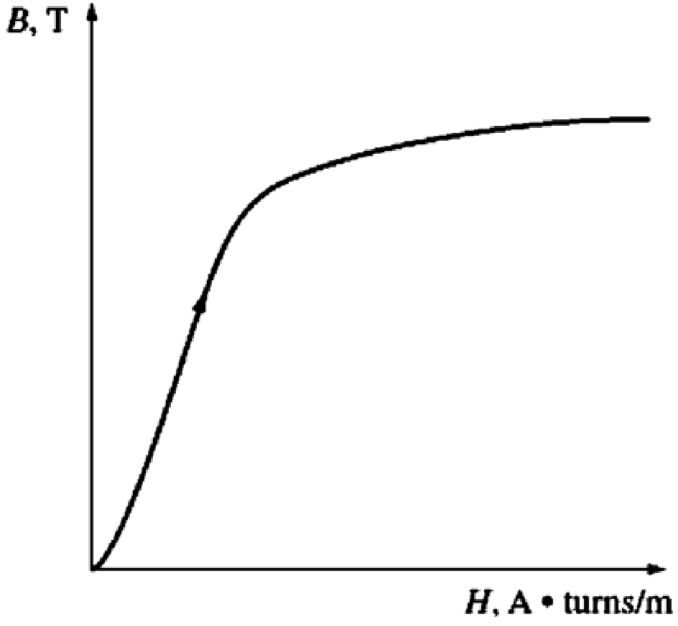


$$\boxed{\mathcal{F} = \phi \mathcal{R}}$$

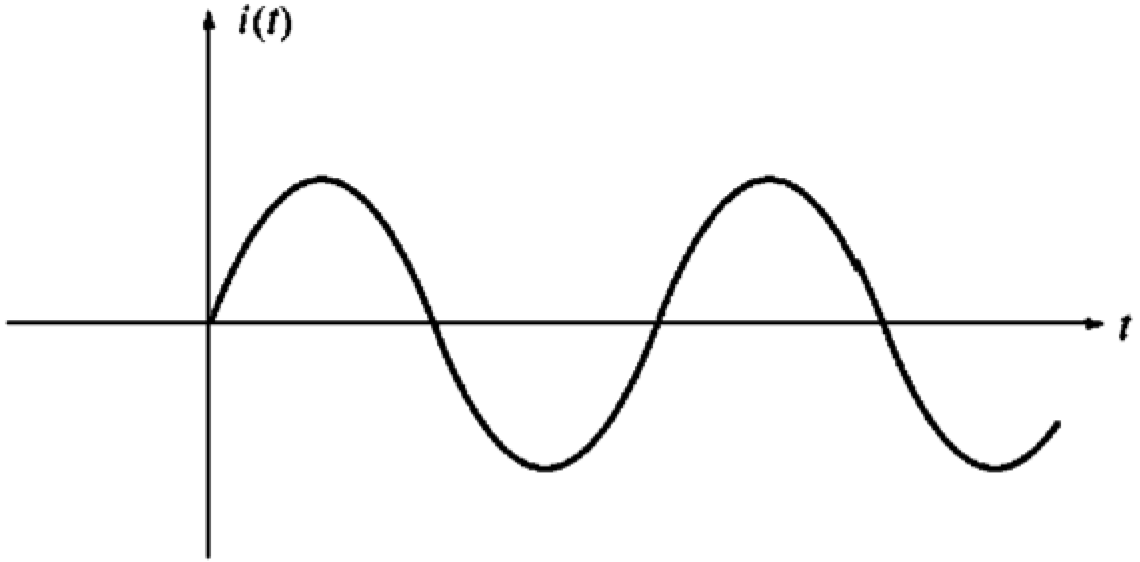
$$H = \frac{Ni}{l_c} = \frac{\mathcal{F}}{l_c}$$

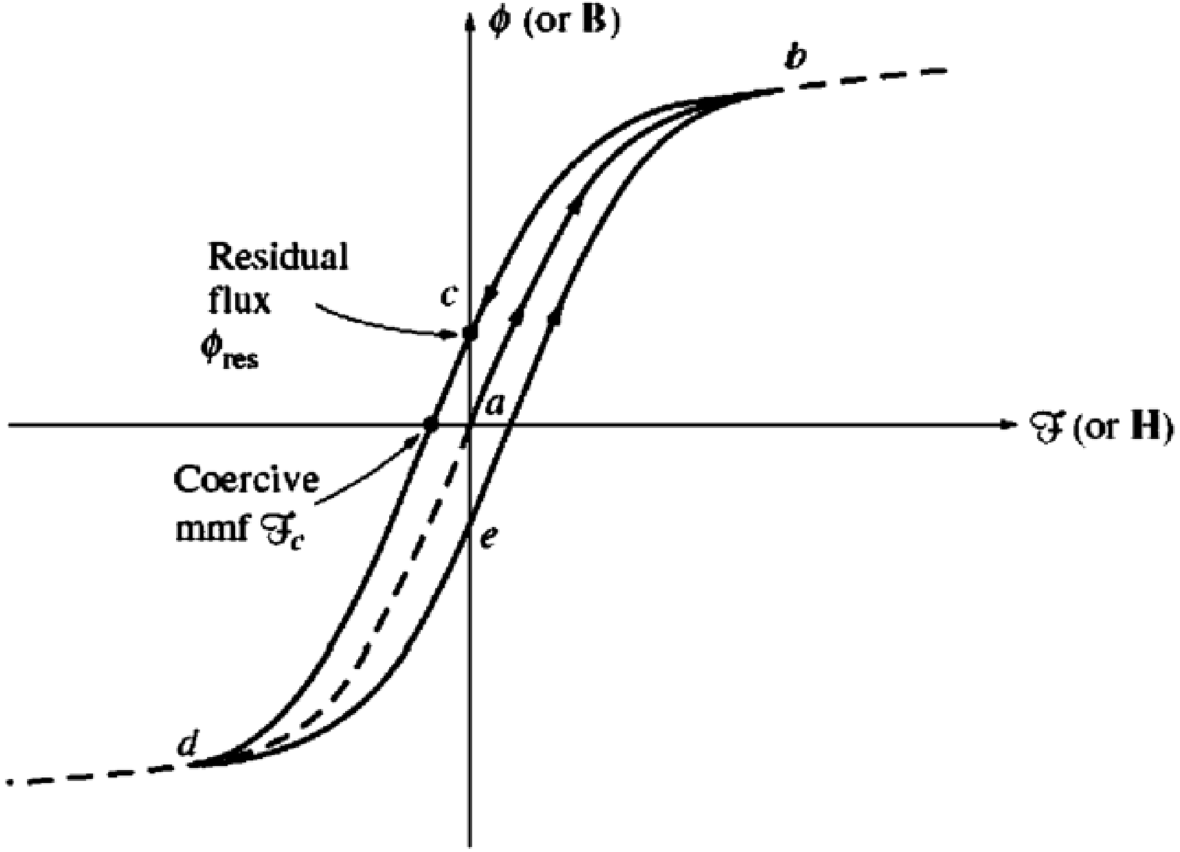
$$\phi = BA$$





The hysteresis loop traced out by the flux in a core when the current i(t) is applied to it.





Example 3: A square magnetic core has a mean path length of 55 cm and a cross-sectional area of 150cm2 . A 200-turn coil of wire is wrapped around one leg of the core. The core is made of a material having the magnetization curve shown in the following Figure.

(a) How much current is required to produce 0.012 Wb of flux in the core?

(b) What is the core's relative permeability at that current level?

(c) What is its reluctance?